

CSE4203: Computer Graphics
Lecture – 5 (part - A)
Viewing

Outline

- Image-order and object-render rendering
- Viewing transformation
- Viewport transformation
- Orthographic projection transformation

Rendering Techniques (1/2)

- One of the basic tasks of computer graphics is rendering 3D objects:
 - taking a scene, or model, composed of many geometric objects arranged in 3D space
 - producing a 2D image that shows the objects as viewed
 - from a particular viewpoint.

Rendering Techniques (2/2)

1. Image-order rendering: iterate over the pixels in the image to be produced, rather than the elements in the scene to be rendered.
2. object-order rendering: that iterate over the elements in the scene to be rendered, rather than the pixels in the image to be produced.

Image-order Rendering (1/2)

- Image-order rendering:
 - *Ray-tracing:*
For each pixel is considered in turn,
 - All the objects that influence it are found
 - and the pixel value is computed.

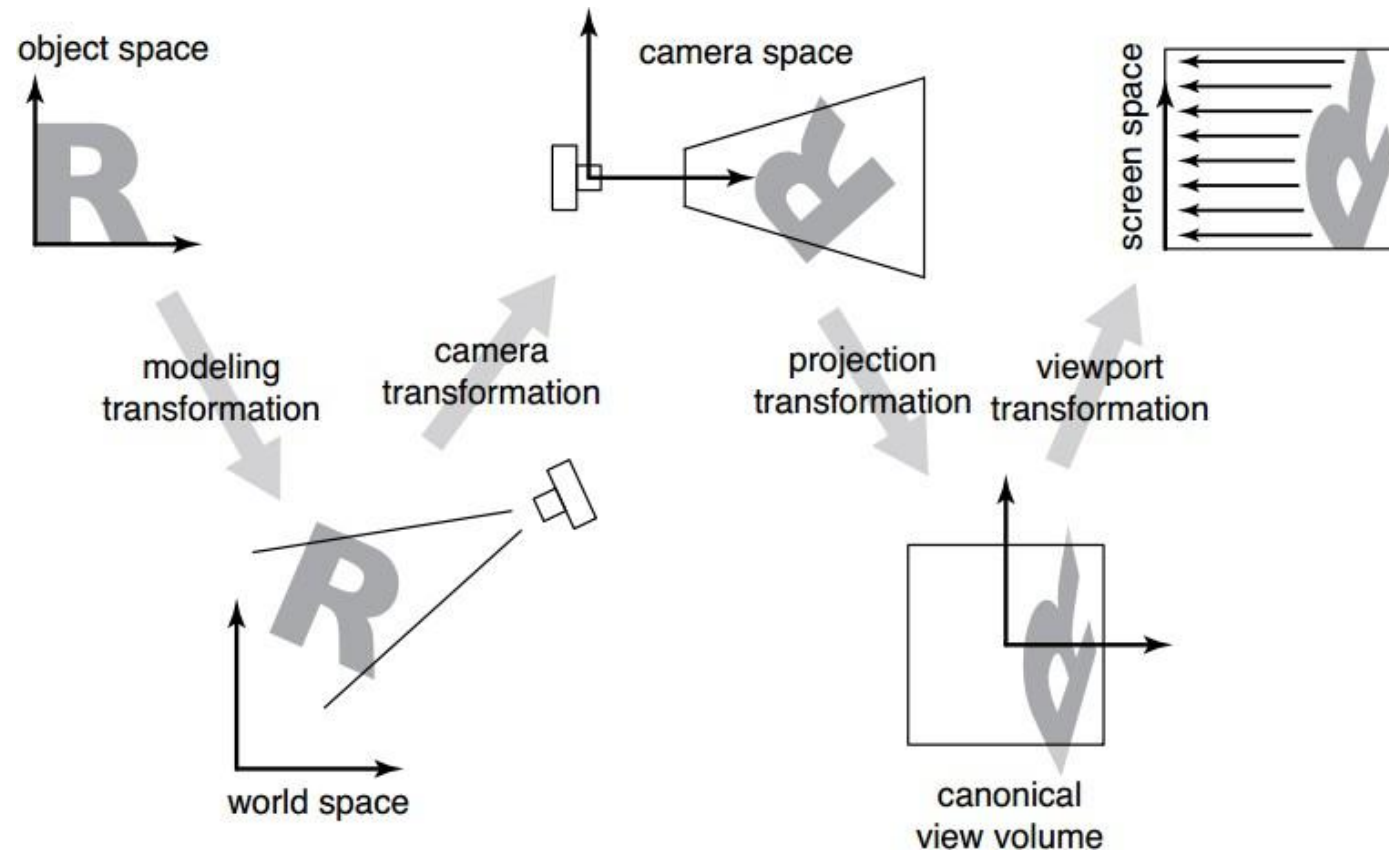
Object-order Rendering (1/2)

- Object-order rendering:
 - *Viewing Transformation:*
For each object is considered in turn,
All the pixels that it influences are found and updated

Object-order Rendering (2/2)

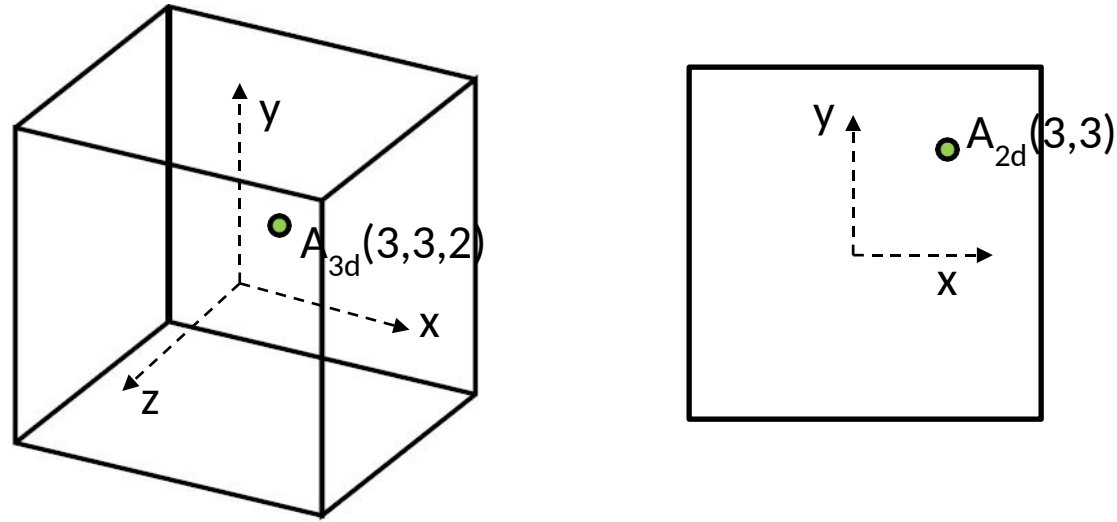
- **Viewing Transformation:**
 - How to use *matrix transformations* to express any parallel or perspective view.
 - These transformations:
 - Project 3D points in the scene (world space) to 2D points in the image (image space)

Viewing Transformation Sequences (1/1)



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>

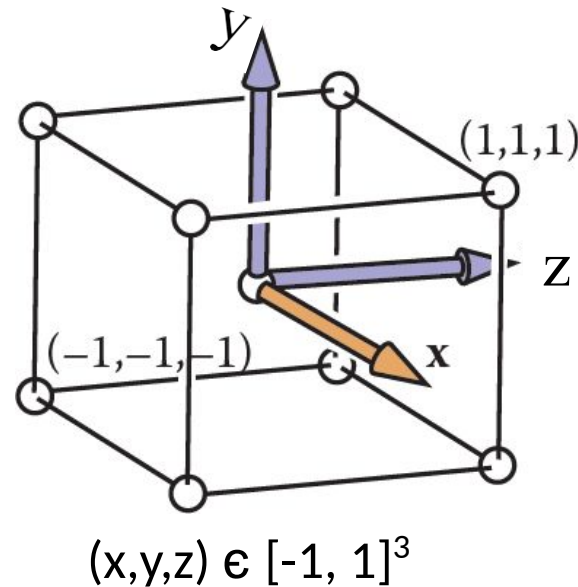
Viewport Transformation (1/19)



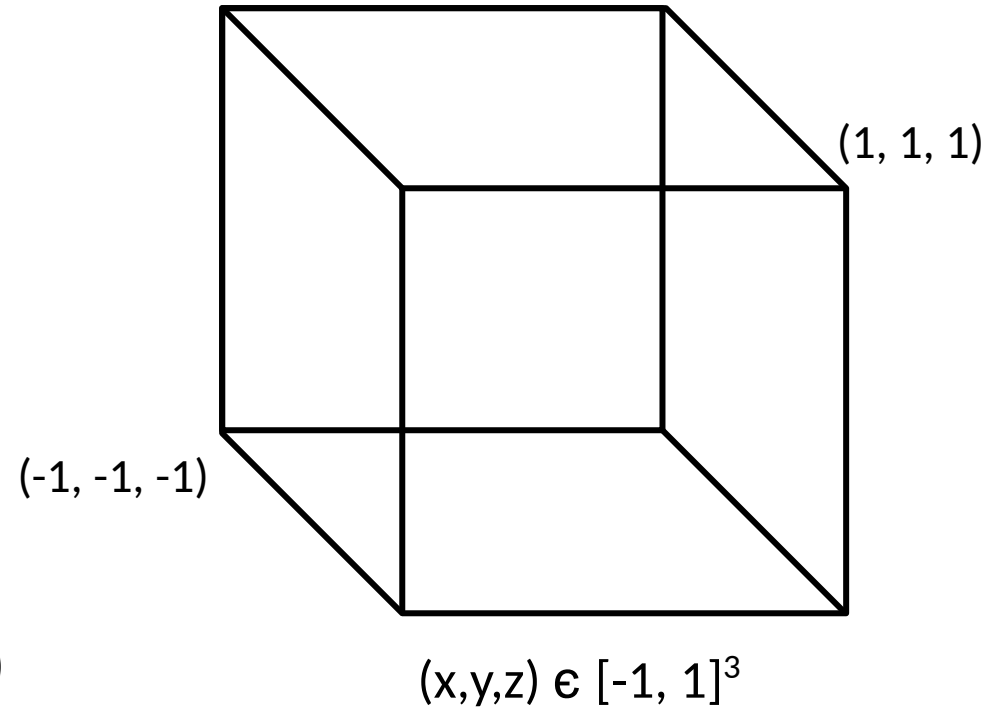
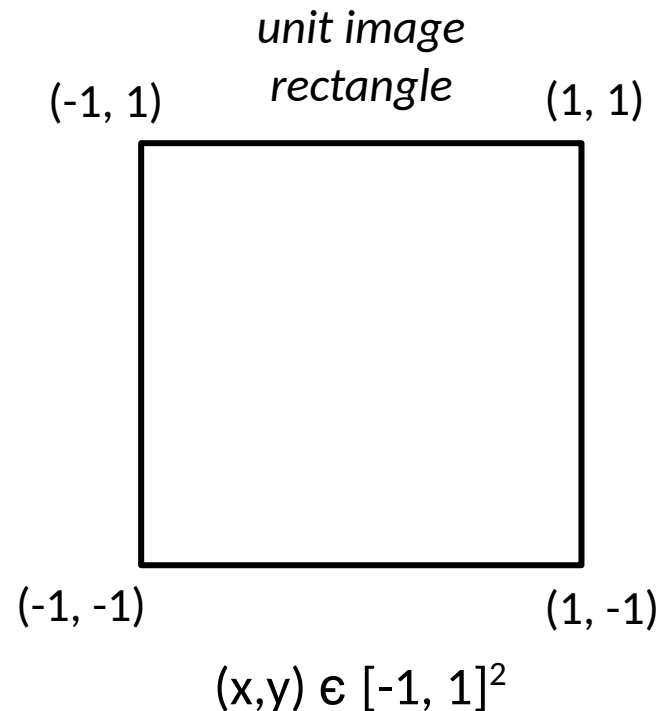
- Projection: Ignoring the z -coordinate

Viewport Transformation (2/19)

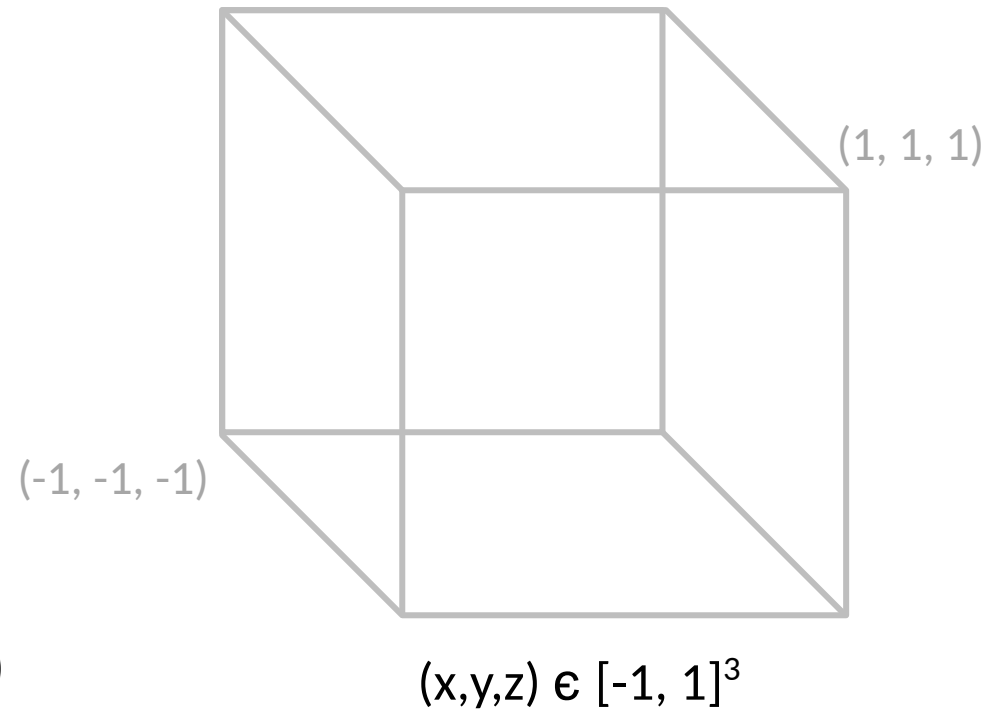
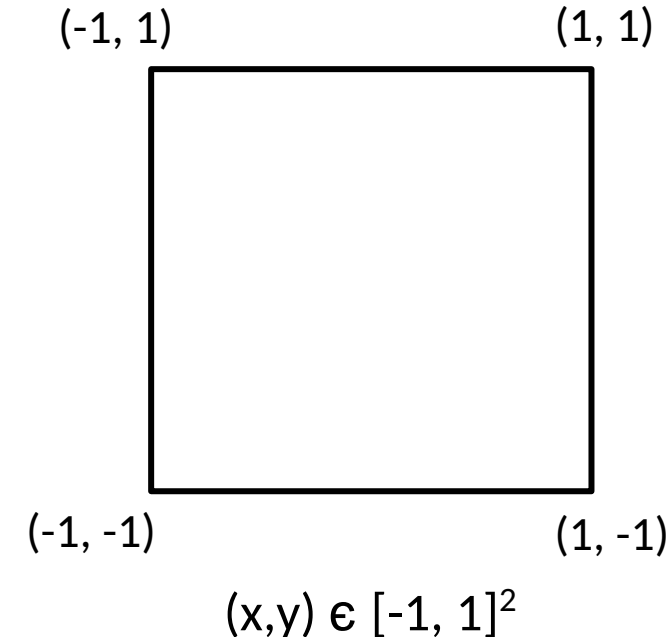
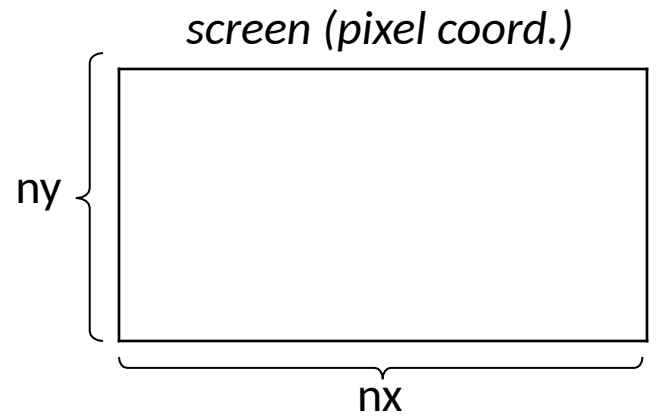
- Canonical View Volume: $(x,y,z) \in [-1, 1]^3$
 - We will assume that the model to be drawn are completely inside the canonical view vol.



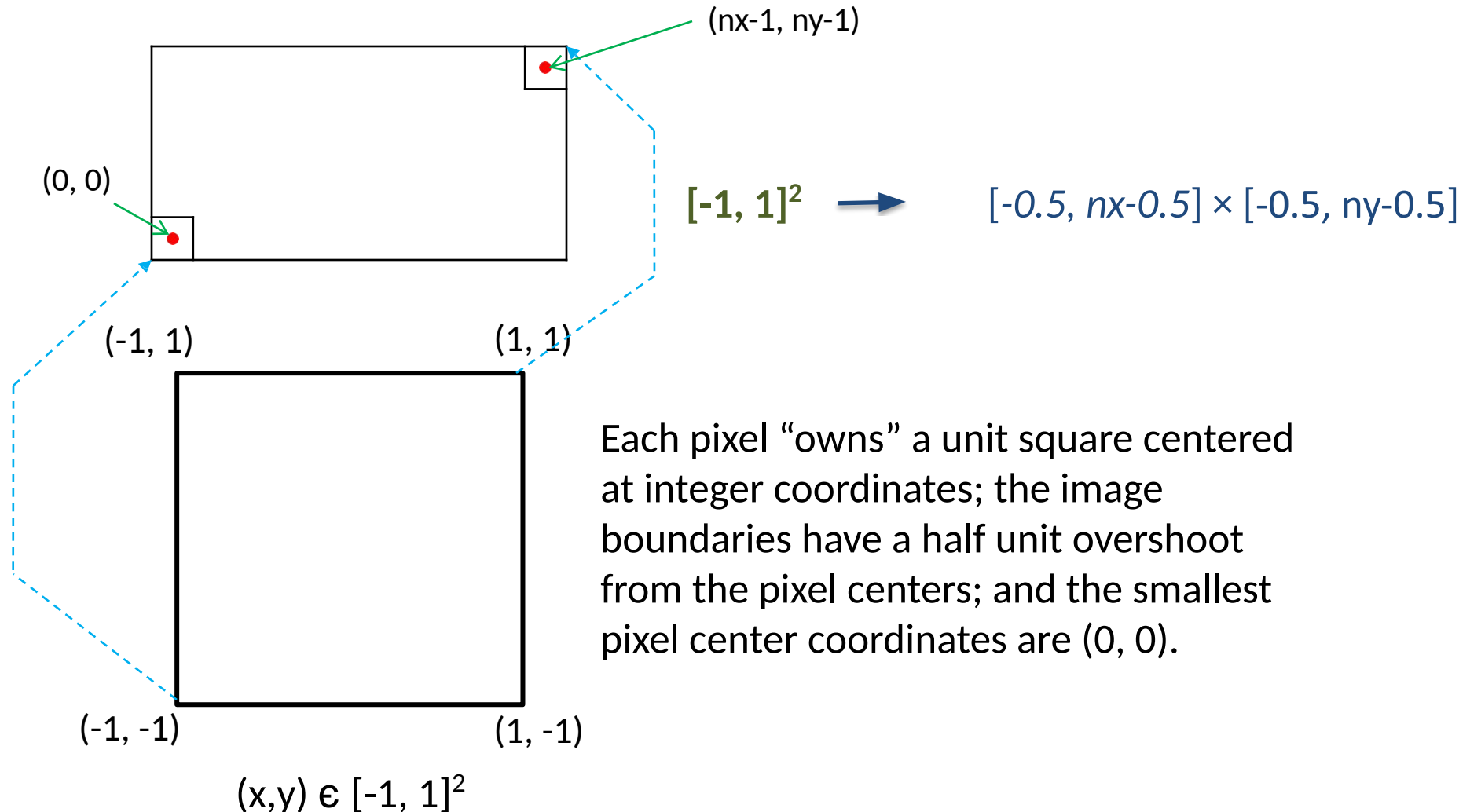
Viewport Transformation (4/19)



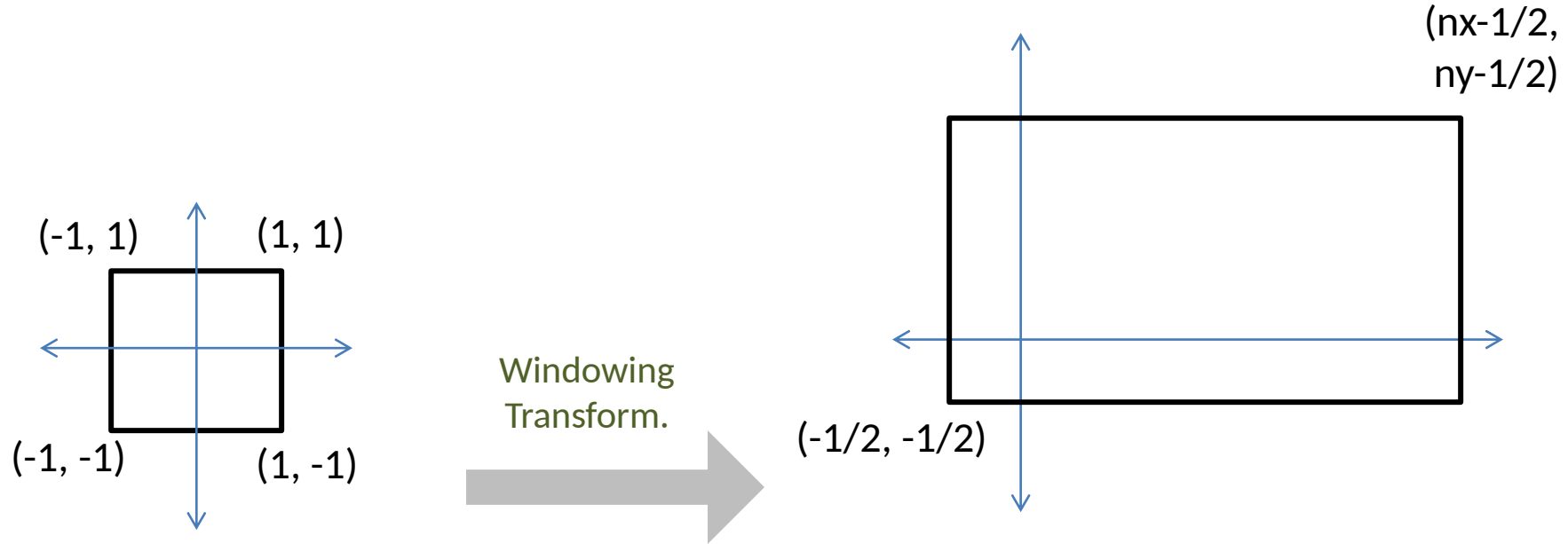
Viewport Transformation (5/19)



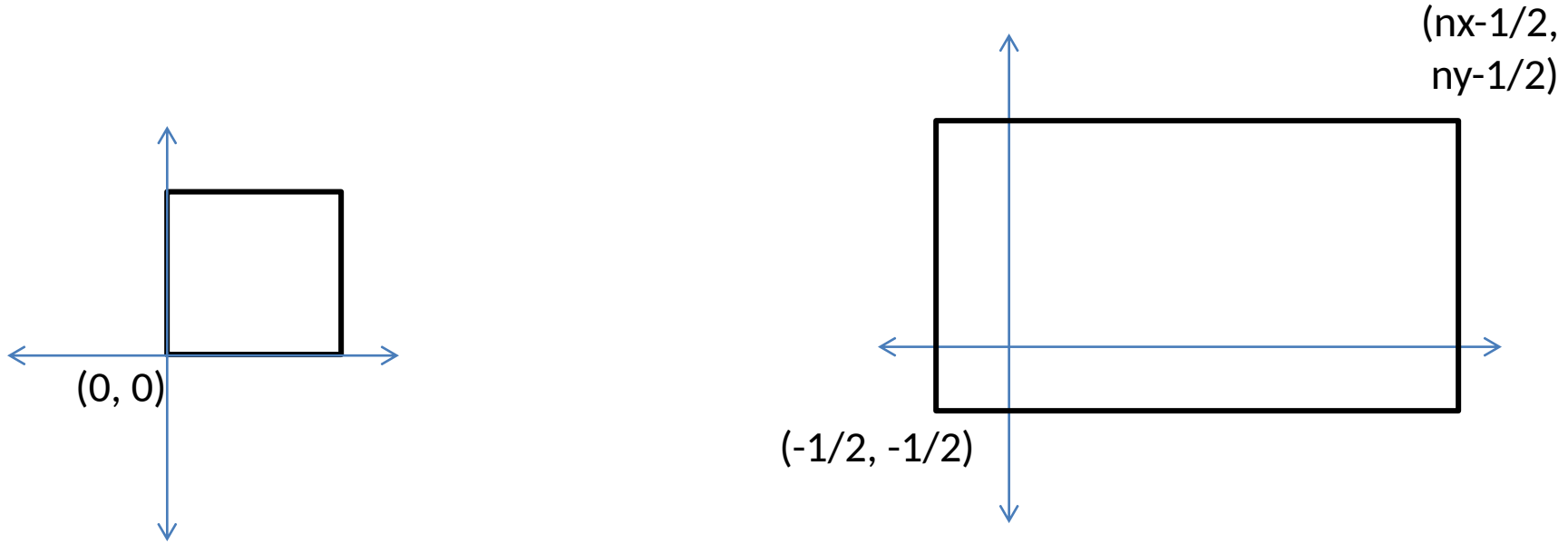
Viewport Transformation (7/19)



Viewport Transformation (8/19)

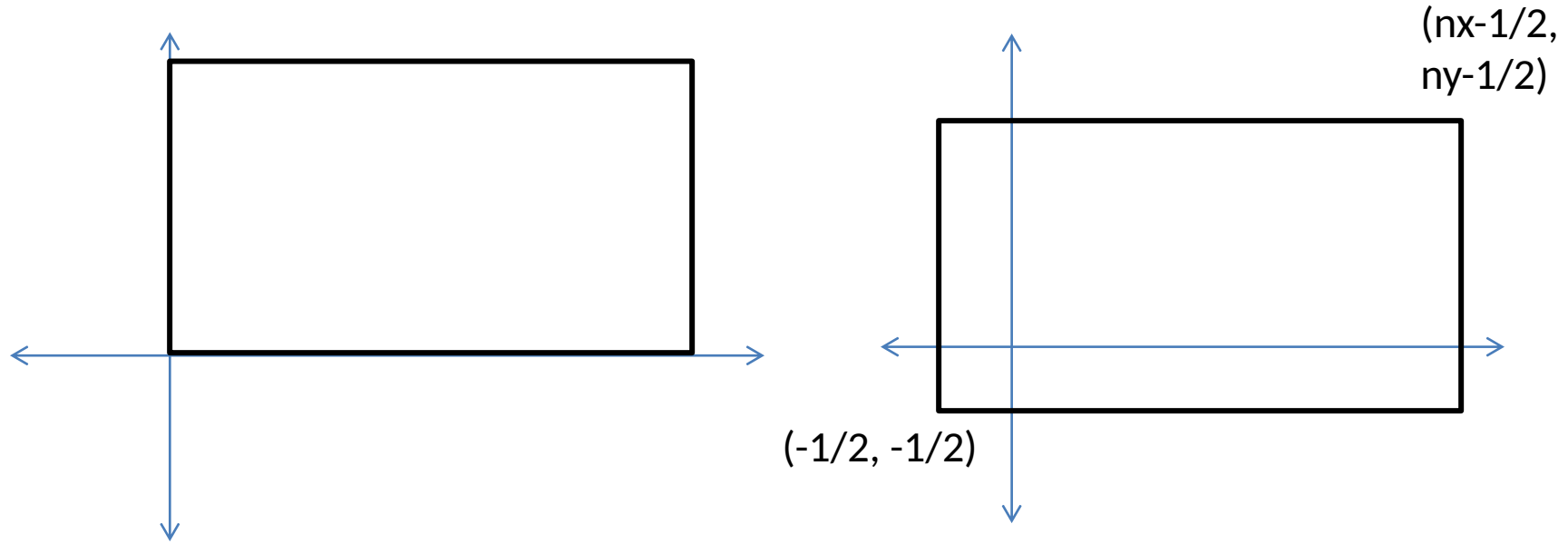


Viewport Transformation (9/19)



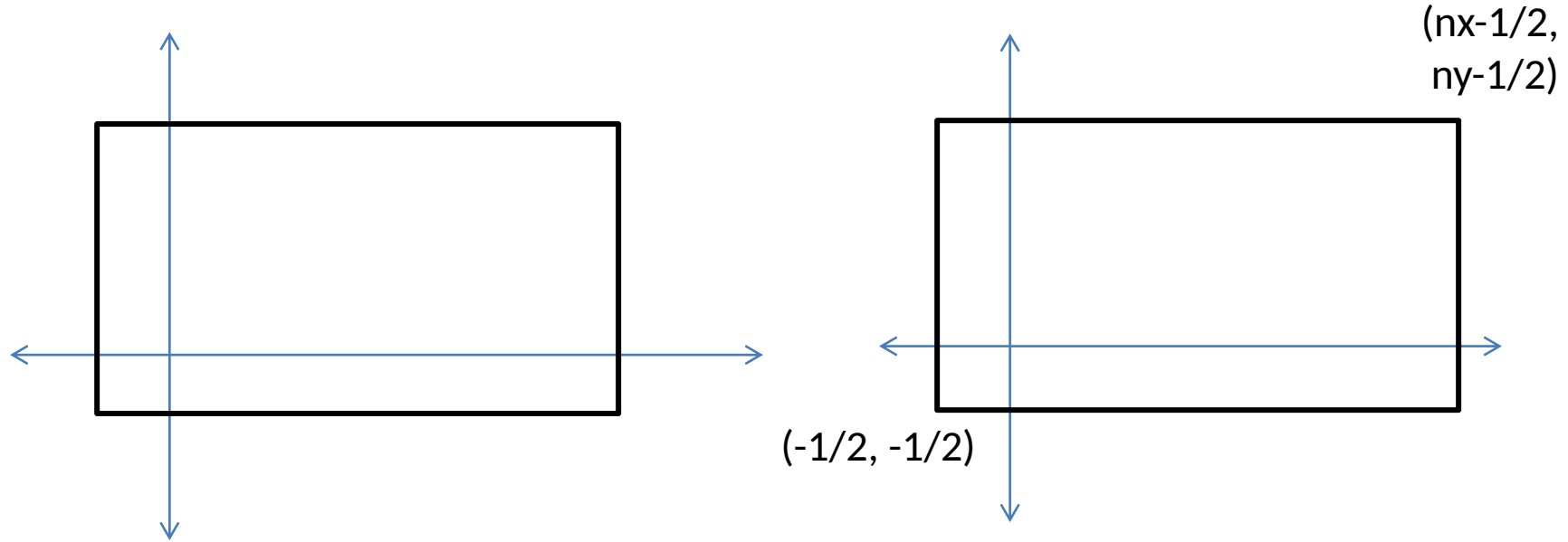
$T(1, 1)$

Viewport Transformation (10/19)



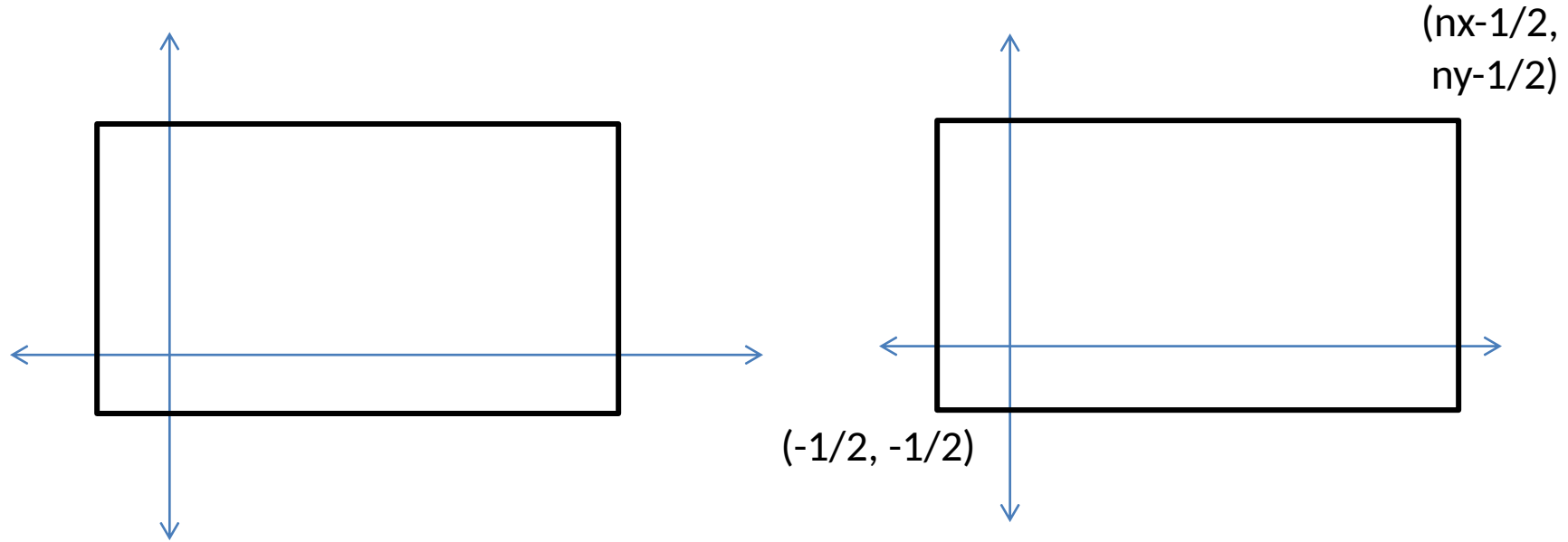
$$T(1, 1) \rightarrow S(nx/2, ny/2)$$

Viewport Transformation (11/19)



$$T(1, 1) \rightarrow S(nx/2, ny/2) \rightarrow T(-1/2, -1/2)$$

Viewport Transformation (12/19)



$$T(1, 1) \rightarrow S(nx/2, ny/2) \rightarrow T(-1/2, -1/2)$$

$$M_{vp} = T(-1/2, -1/2) * S(nx/2, ny/2) * T(1,1)$$

Viewport Transformation (15/19)

$$M_{vp} = T(-1/2, -1/2) * S(n_x/2, n_y/2) * T(1, 1)$$

$$\begin{bmatrix} \frac{n_x}{2} & 0 & \frac{n_x-1}{2} \\ 0 & \frac{n_y}{2} & \frac{n_y-1}{2} \\ 0 & 0 & 1 \end{bmatrix}$$

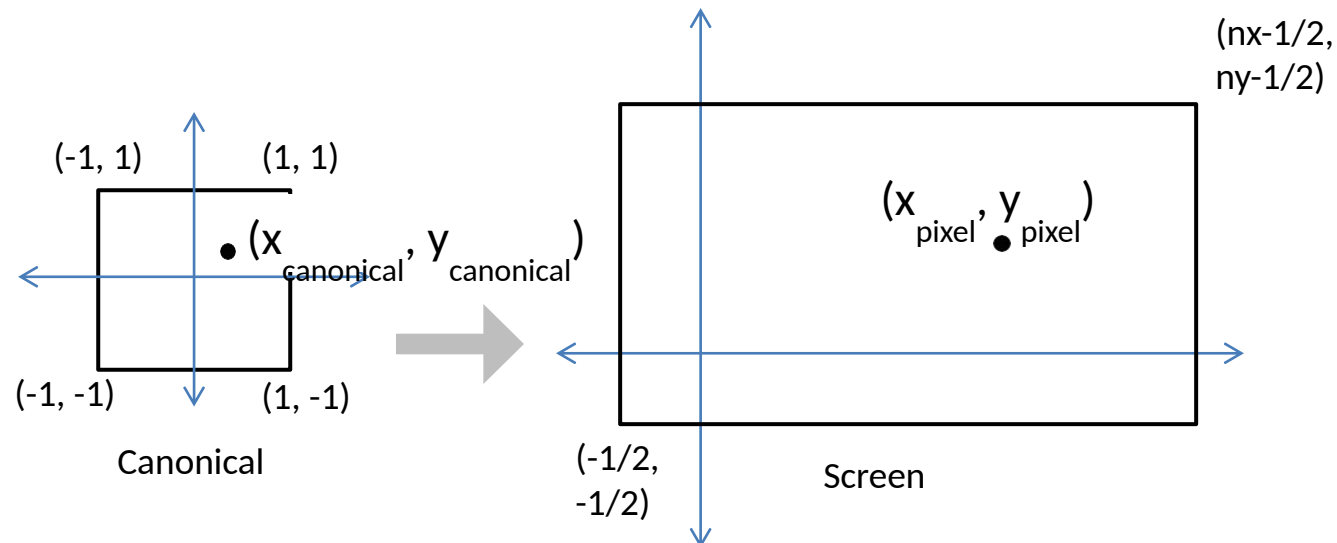
Q: Do matrix
multiplication and check

Viewport Transformation (16/19)

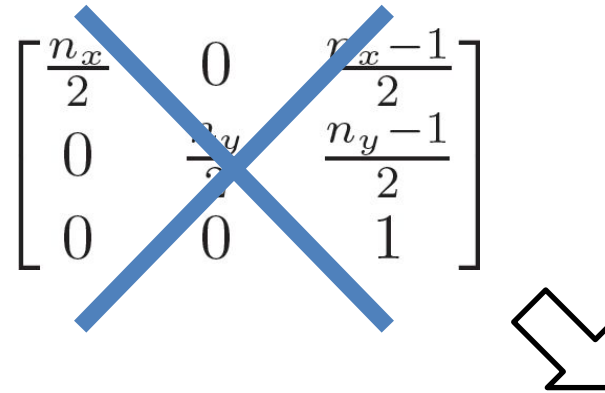
$$M_{vp} = T(-1/2, -1/2) * S(n_x/2, n_y/2) * T(1, 1)$$

This is similar
to windowing
transform.
(Chap - 6)

$$\begin{bmatrix} x_{\text{pixel}} \\ y_{\text{pixel}} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{n_x}{2} & 0 & \frac{n_x-1}{2} \\ 0 & \frac{n_y}{2} & \frac{n_y-1}{2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{\text{canonical}} \\ y_{\text{canonical}} \\ 1 \end{bmatrix}$$



Viewport Transformation (18/19)

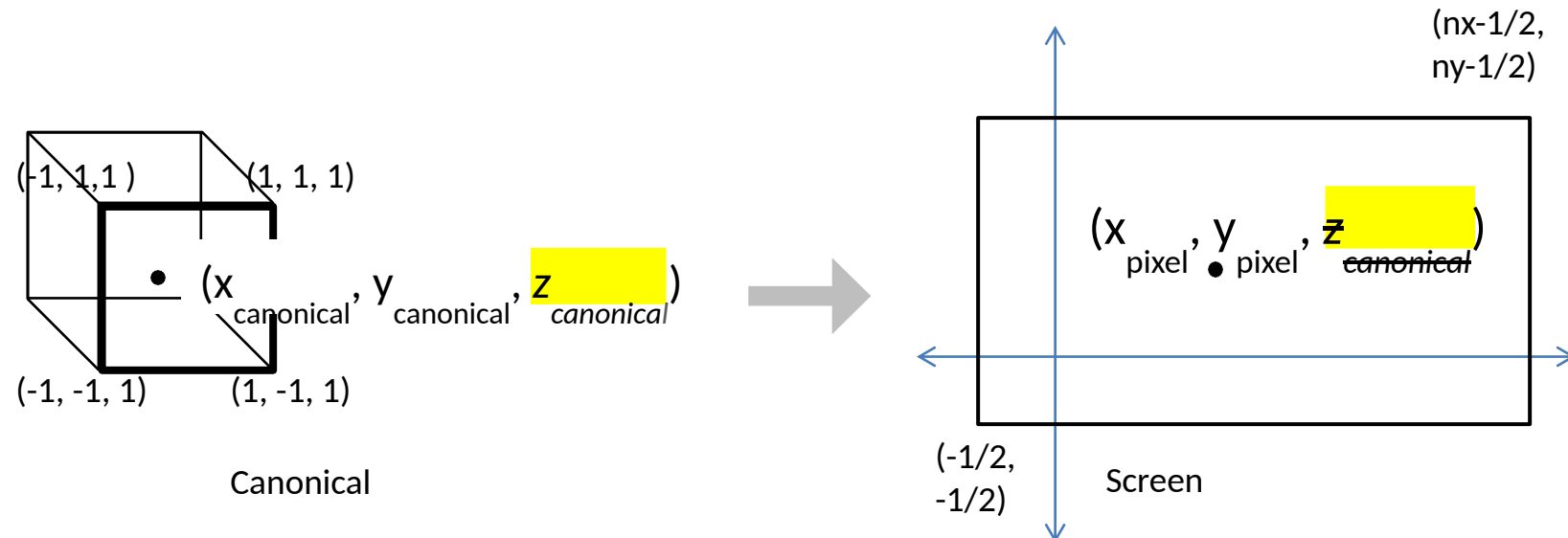

$$\begin{bmatrix} \frac{n_x}{2} & 0 & \frac{n_x-1}{2} \\ 0 & \frac{n_y}{2} & \frac{n_y-1}{2} \\ 0 & 0 & 1 \end{bmatrix}$$


$$M_{vp} = \begin{bmatrix} \frac{n_x}{2} & 0 & 0 & \frac{n_x-1}{2} \\ 0 & \frac{n_y}{2} & 0 & \frac{n_y-1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Viewport Transformation (19/19)

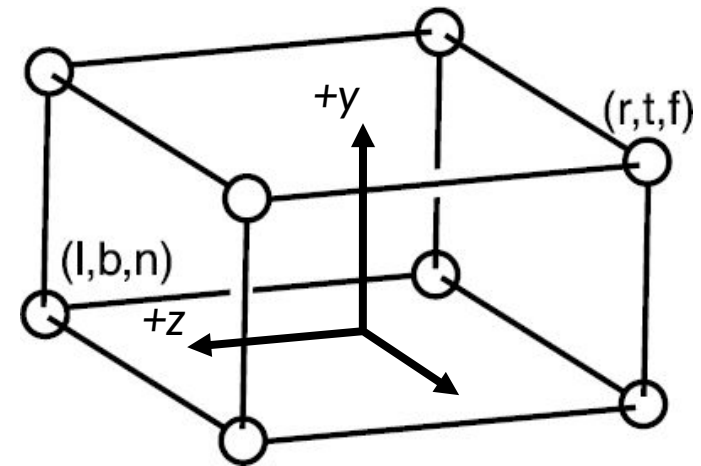
[carry along the z-coordinate without changing it]

$$\begin{bmatrix} x_{\text{pixel}} \\ y_{\text{pixel}} \\ z_{\text{canonical}} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{n_x}{2} & 0 & 0 & \frac{n_x-1}{2} \\ 0 & \frac{n_y}{2} & 0 & \frac{n_y-1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{\text{canonical}} \\ y_{\text{canonical}} \\ z_{\text{canonical}} \\ 1 \end{bmatrix}$$



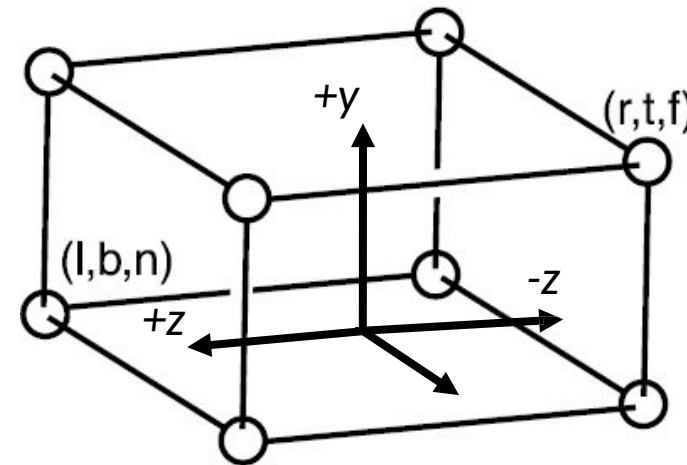
Orthographic Projection Transformation (1/1)

- What if we want to render geometry in some region other than canonical view vol.?



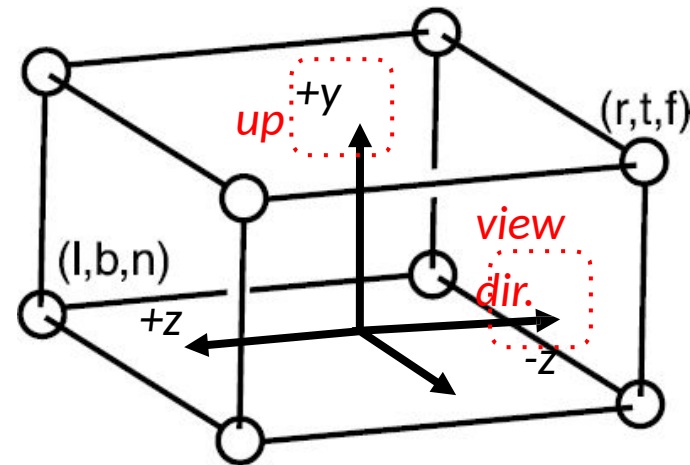
Orthographic View Volume (1/3)

- We'll name the coordinates of its sides so that the view volume is $[l, r] \times [b, t] \times [f, n]$
 - View direction: *looking along $-z$*
 - Orientation: *$+y$ up*
- $x = l \equiv$ left plane,
- $x = r \equiv$ right plane,
- $y = b \equiv$ bottom plane,
- $y = t \equiv$ top plane,
- $z = n \equiv$ near plane,
- $z = f \equiv$ far plane.



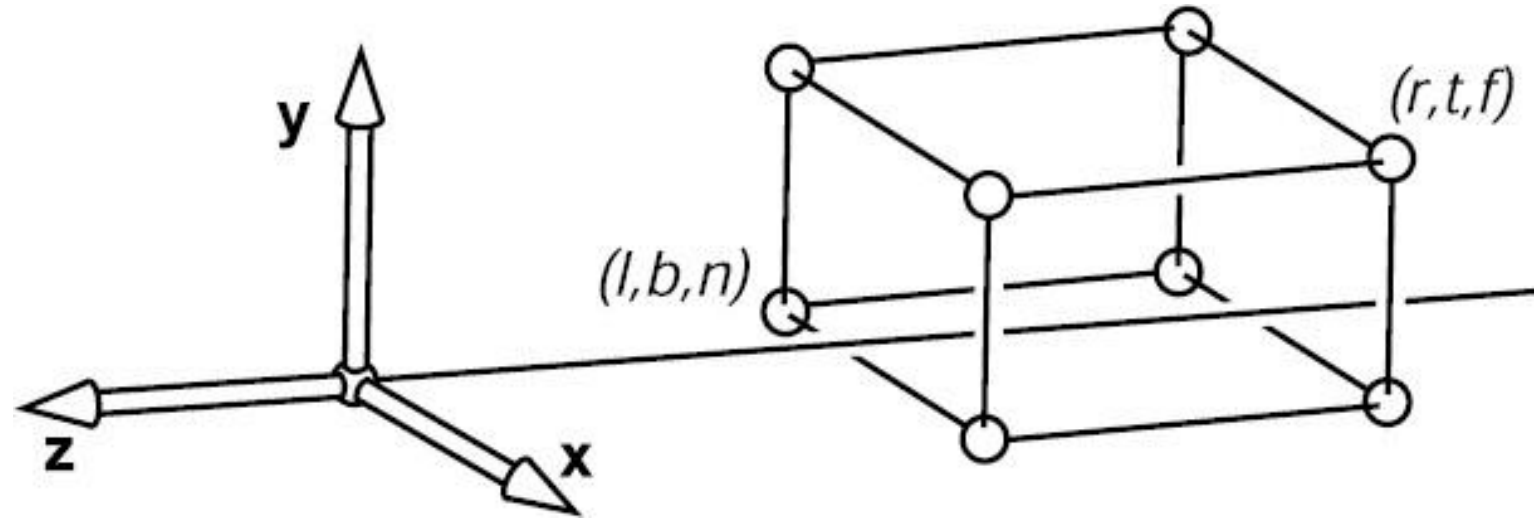
Orthographic View Volume (2/3)

- Looking along the *minus z-axis* with his head pointing in the *positive y-direction*.
 - View direction: *looking along -z*
 - Orientation: *+y up*
- *But, this is unintuitive!*



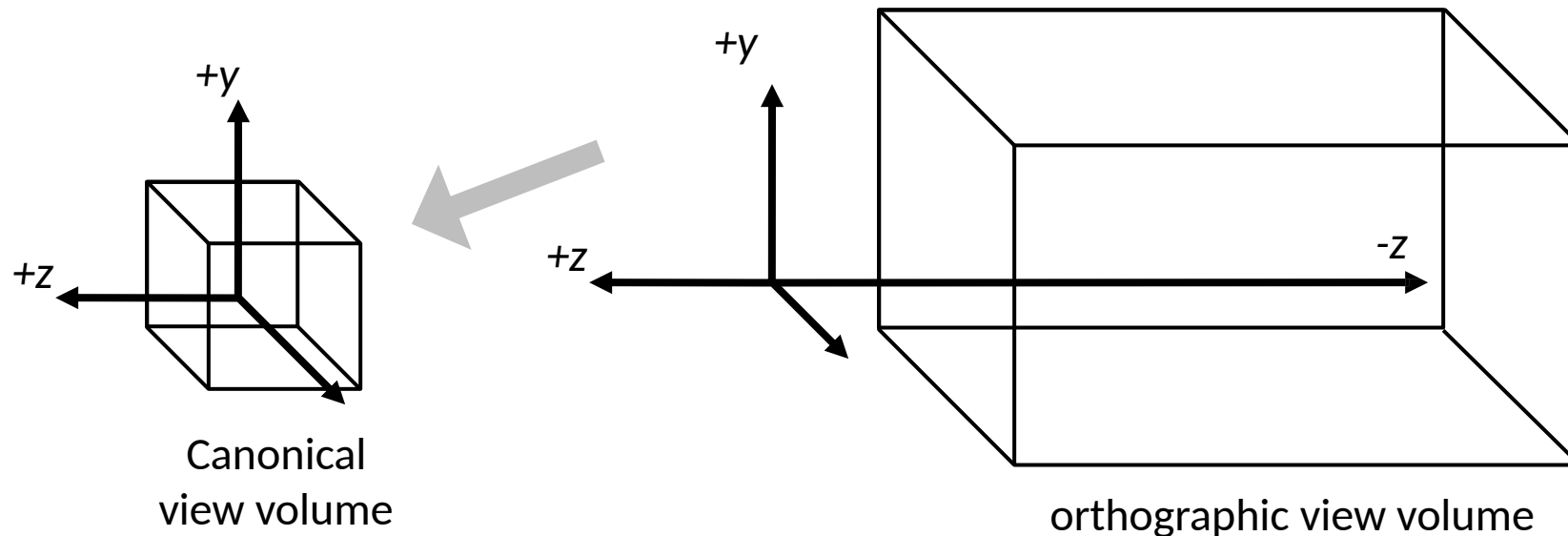
Orthographic View Volume (3/3)

- If entire orthographic view volume has **negative z** then $n > f$.
 - $z = n$ plane is closer



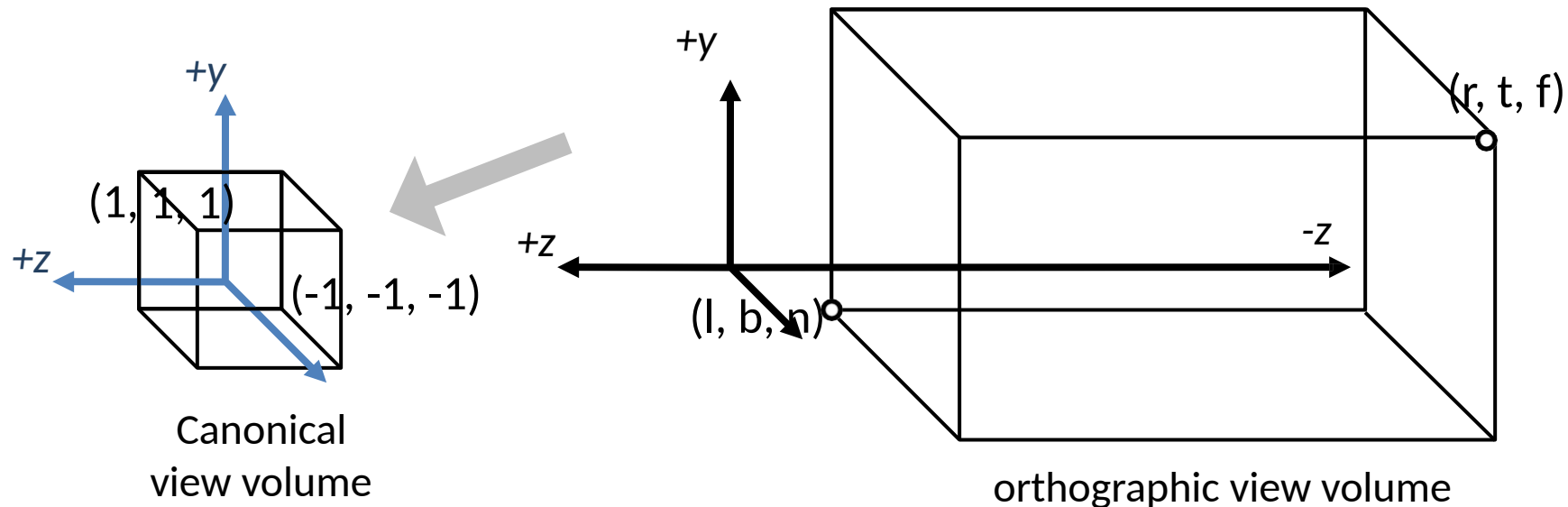
Orthographic to Canonical View Volume (1/3)

- Transform from orthographic view volume to the canonical view volume
 - We need to apply *windowing transformation* (just like before!)



Orthographic to Canonical View Volume (2/3)

$$\mathbf{M}_{\text{orth}} = \begin{bmatrix} \frac{2}{r-l} & 0 & 0 & -\frac{r+l}{r-l} \\ 0 & \frac{2}{t-b} & 0 & -\frac{t+b}{t-b} \\ 0 & 0 & \frac{2}{n-f} & -\frac{n+f}{n-f} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

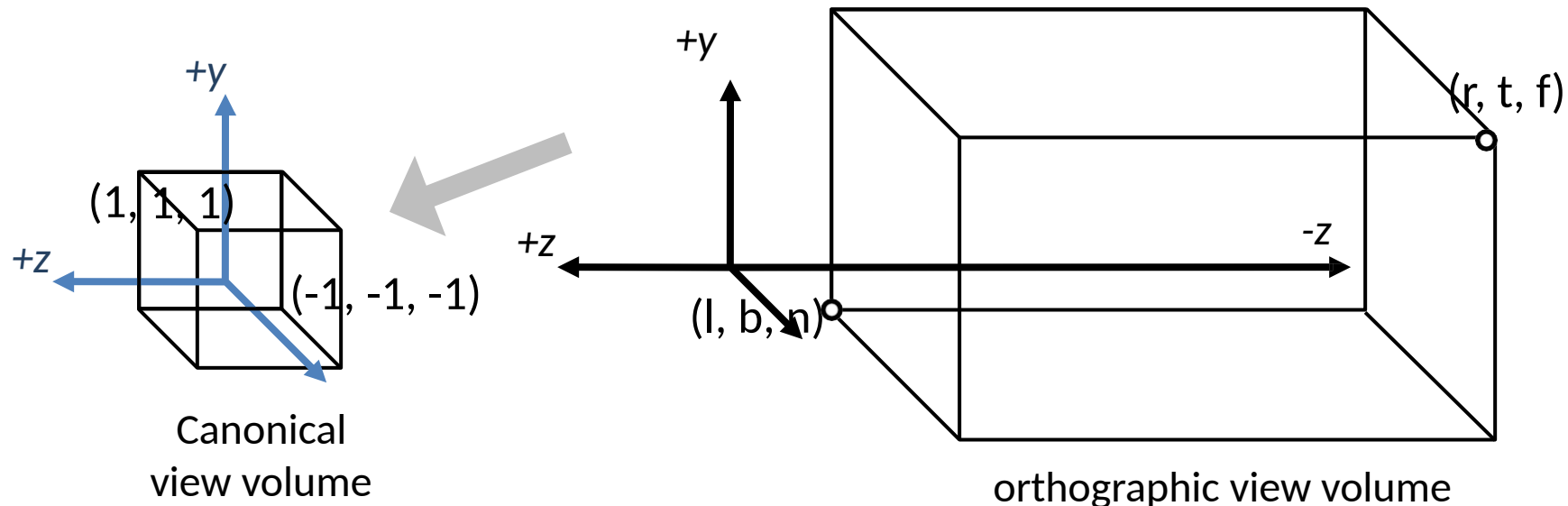


Orthographic to Canonical View Volume (3/3)

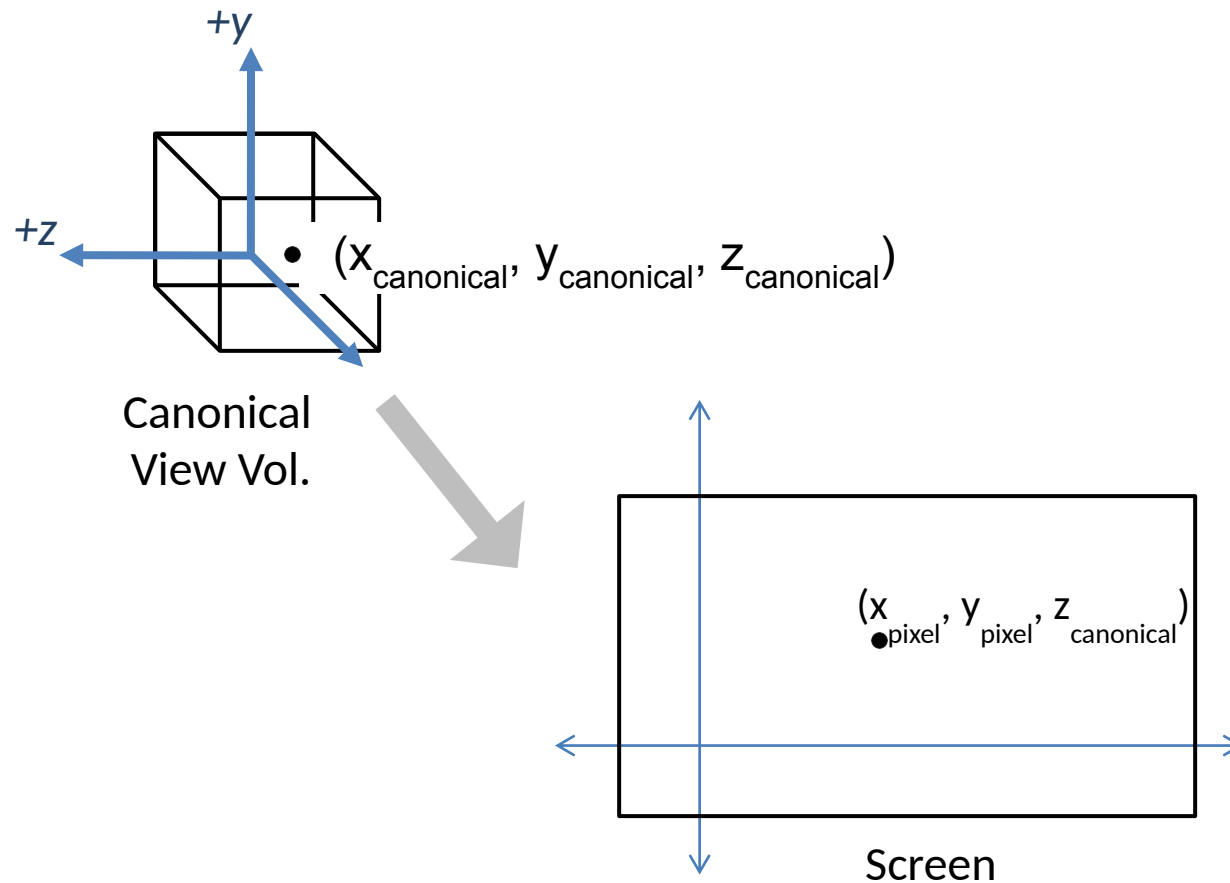
$$\mathbf{M}_{\text{orth}} = \begin{bmatrix} \frac{2}{r-l} & 0 & 0 & -\frac{r+l}{r-l} \\ 0 & \frac{2}{t-b} & 0 & -\frac{t+b}{t-b} \\ 0 & 0 & \frac{2}{n-f} & -\frac{n+f}{n-f} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Q: How can we get this matrix?

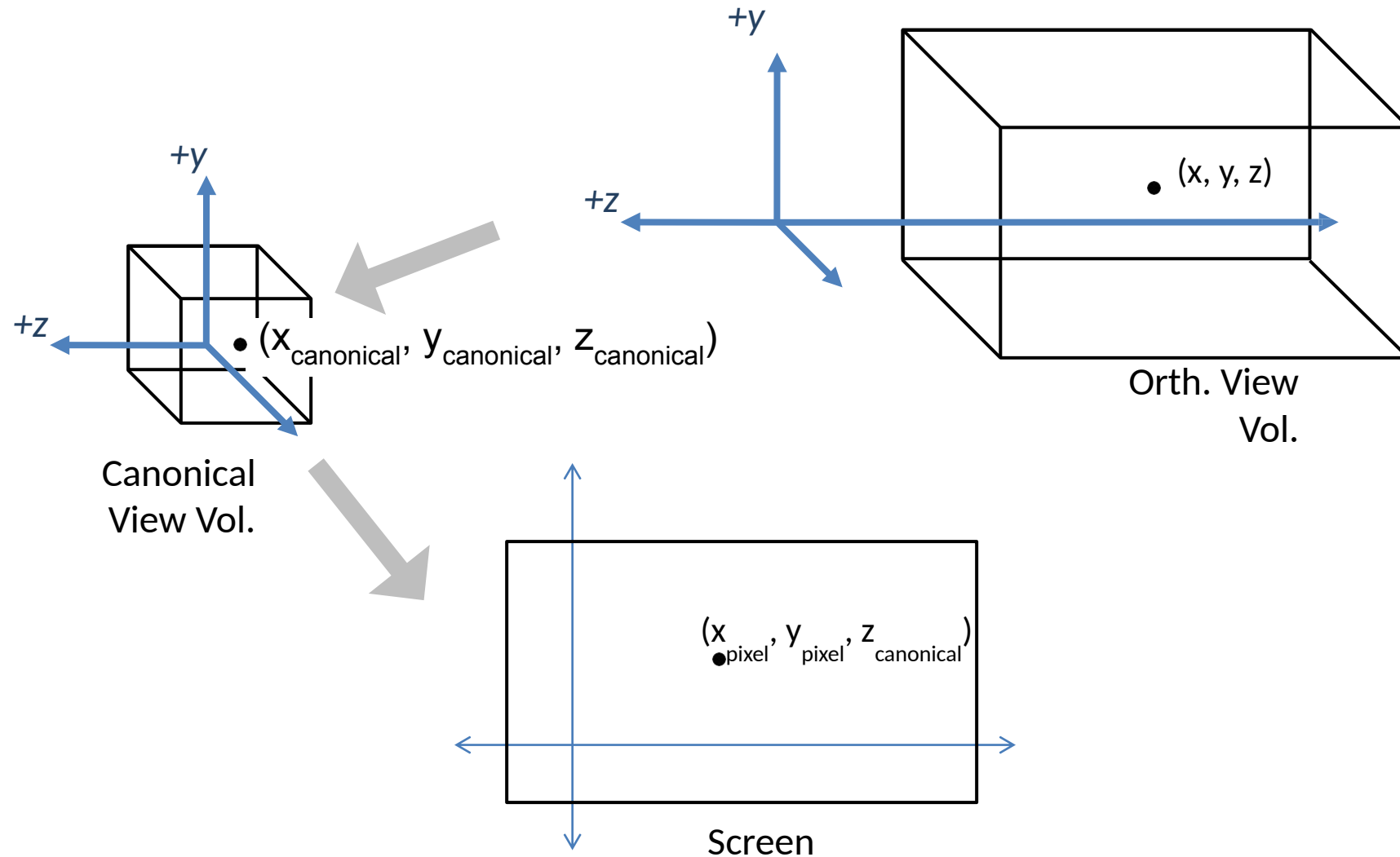
Help: Chap 6 (Windowing Transformation) and M_{vp}



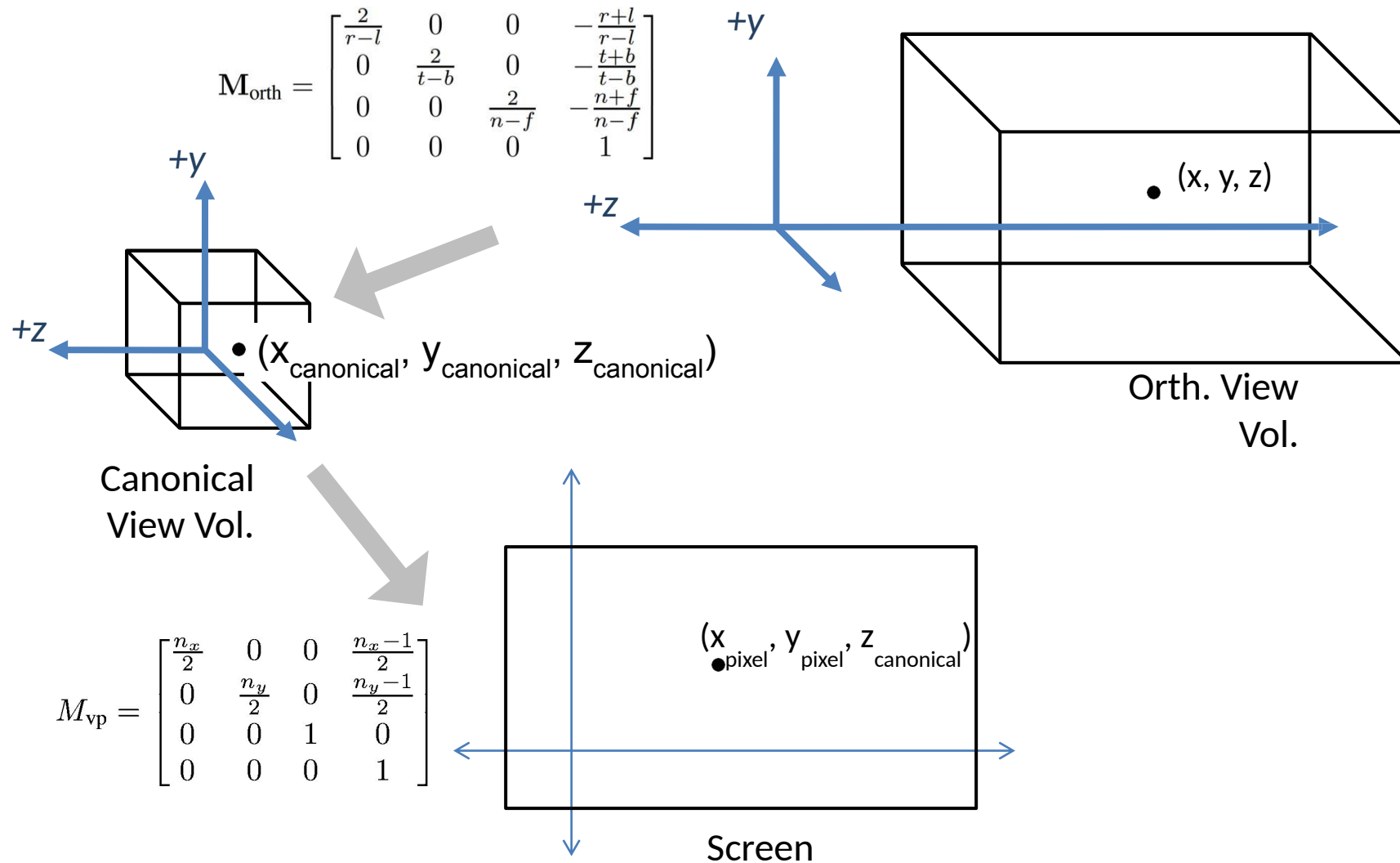
Orthographic $\rightarrow$ Canonical $\rightarrow$ Screen (1/5)



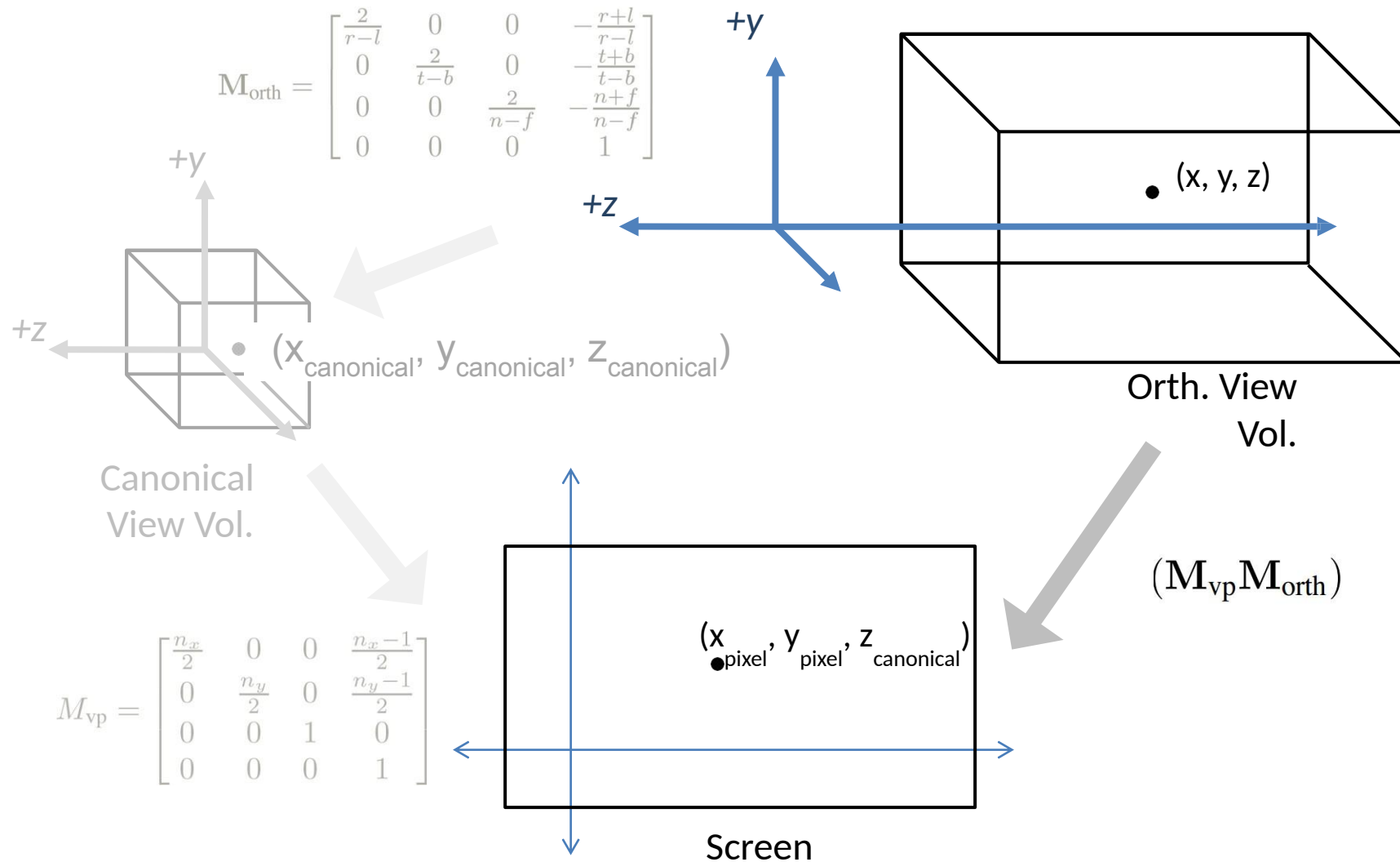
Orthographic → Canonical → Screen (2/5)



Orthographic → Canonical → Screen (3/5)

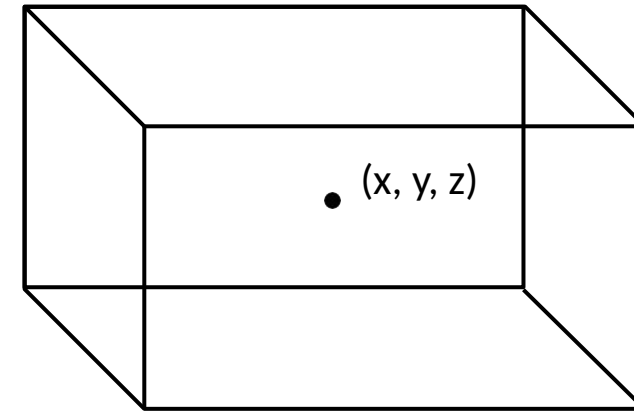


Orthographic → Canonical → Screen (4/5)

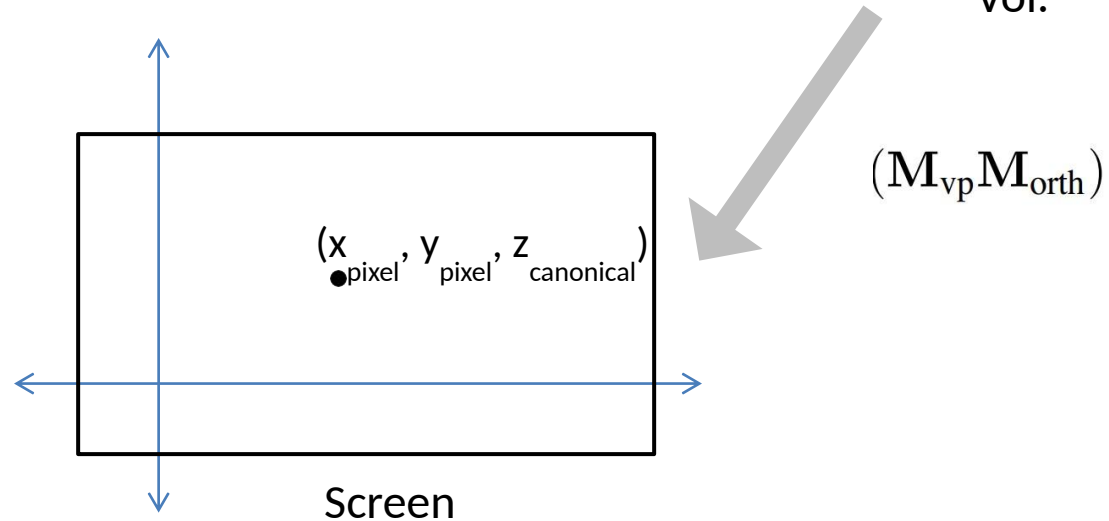


Orthographic → Canonical → Screen (5/5)

$$\begin{bmatrix} x_{\text{pixel}} \\ y_{\text{pixel}} \\ z_{\text{canonical}} \\ 1 \end{bmatrix} = (\mathbf{M}_{\text{vp}} \mathbf{M}_{\text{orth}}) \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



Orth. View
Vol.



Code: Orthographic to Screen (1/1)

Drawing many 3D lines with endpoints a_i and b_i :

```
Construct  $M_{vp}$ 
```

```
Construct  $M_{orth}$ 
```

```
 $M = M_{vp} * M_{orth}$ 
```

```
for each line segment  $(a_i, b_i)$  do:
```

```
     $p = M * a_i$ 
```

```
     $q = M * b_i$ 
```

```
    drawline  $(x_p, y_p, x_q, y_q)$ 
```

Practice Problem - 1

Transform a 3D line AB from an *orthographic view volume* to a *viewport* of size 128 x 96. Vertices of the line are A(-1, -3, -5) and B(2, 4, -6). The orthographic view volume has the following setup:

$$l = -4, r = 4, b = -4, t = 4, n = -4, f = -8$$

You must -

- a. Determine the transformation matrix M .
- b. Multiply M with the vertices of the line and determine the position of vertices on viewport.

Practice Problem – 1 (Sol.)

$$M_{vp} = \begin{bmatrix} \frac{n_x}{2} & 0 & 0 & \frac{n_x-1}{2} \\ 0 & \frac{n_y}{2} & 0 & \frac{n_y-1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$n_x = 128$ $n_y = 96$

$$M_{orth} = \begin{bmatrix} \frac{2}{r-l} & 0 & 0 & -\frac{r+l}{r-l} \\ 0 & \frac{2}{t-b} & 0 & -\frac{t+b}{t-b} \\ 0 & 0 & \frac{2}{n-f} & -\frac{n+f}{n-f} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$l = -4, r = 4,$ $b = -4, t = 4,$ $n = -4, f = -8$
--

Practice Problem – 1 (Sol.)

$$M = M_{vp} * M_{orth}$$

$$M = \begin{bmatrix} \frac{2}{r-l} & 0 & 0 & -\frac{r+l}{r-l} \\ 0 & \frac{2}{t-b} & 0 & -\frac{t+b}{t-b} \\ 0 & 0 & \frac{2}{n-f} & -\frac{n+f}{n-f} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{n_x}{2} & 0 & 0 & \frac{n_x-1}{2} \\ 0 & \frac{n_y}{2} & 0 & \frac{n_y-1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A' = M A$$

$$B' = M B$$

Further Reading

- Fundamentals of Computer Graphics, 4th Edition - Chapter 7